



Indian Iron & Steel Sector Skill Council

QUALIFICATIONS PACK - OCCUPATIONAL STANDARDS FOR IRON & STEEL INDUSTRY

What are Occupational Standards (OS)?

- OS describe what individuals need to do, know and understand in order to carry out a particular job role or function
- OS are performance standards that individuals must achieve when carrying out functions in the workplace, together with specifications of the underpinning knowledge and understanding



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Introduction

Qualifications Pack - **Battery Operator**

SECTOR: Iron & Steel

SUB-SECTOR: Steel

REFERENCE ID: ISC/Q0201

ALIGNED TO: NCO-2004/NIL

Title of Job: The job is all about carrying out all enabling activities which are integral part of coke making in a slotted coke oven battery for producing metallurgical coke.

Personal Attributes: This job requires the individual to work independently as well as in teams. He should be physically fit, not having colour blindness, having analytical skills, problem solving attitude, high concentration levels and willingness to work in a factory environment.



Job Details

Qualifications Pack Code	ISC/Q0201		
Job Role	Battery Operator		
Credits(NSQF)	TBD	Version number	1.0
Industry	Iron & Steel	Drafted on	12/11/2014
Sub-sector	Steel	Last reviewed on	25/03/2015
Occupation	Coke Making	Next review date	25/03/2016
NSQC Clearance on	20/07/2015		

Job Role	Battery Operator
Role Description	The job holder is responsible for general cleaning, housekeeping, battery machine cleaning, preparatory cleaning, technological cleaning of gas carrying equipments, cleaning, sealing and regulation of all sealing surfaces. Operating pollution control equipments, preparing ovens before pushing/charging and normalising after pushing/charging.
NSQF level	5
Minimum Educational Qualifications	ITI Pass
Maximum Educational Qualifications	Diploma Pass
Training (Suggested but not mandatory)	• Induction training for coke making process including oven preparation for charging and pushing • 6 months on job training (mandatory) • Need based package training for PLC, Hydraulics and computer orientation • Job specific safety training • Awareness programme for pollution control
Minimum Job Entry Age	18 years
Experience	• In lieu of minimum qualification the incumbent should



	have passed / qualified QP of Control room operator/Process operator with 10 th pass
Occupational Standards (OS)	Compulsory: ISC/N0201: Carry out all enabling jobs pertaining to coke production ISC/N0202: Prepare and carry out charging operation ISC/N0203: Prepare and carry out pushing operation ISC/N0008: Use basic health and safety practices at the workplace ISC/N0096: Works effectively with others and manage team members Optional: N/A
Performance Criteria	As described in the relevant NOS units



Definitions

Keywords /Terms	Description
Core Skills/Generic Skills	Core Skills or Generic Skills are a group of skills that are key to learning and working in today's world. These skills are typically needed in any work environment. In the context of the NOS, these include communication related skills that are applicable to most job roles.
Function	Function is an activity necessary for achieving the key purpose of the sector, occupation, or area of work, which can be carried out by a person or a group of persons. Functions are identified through functional analysis and form the basis of NOS.
Job role	Job role defines a unique set of functions that together form a unique employment opportunity in an organization.
Knowledge and Understanding	Knowledge and Understanding are statements which together specify the technical, generic, professional and organizational specific knowledge that an individual needs in order to perform to the required standard.
National Occupational Standards (NOS)	NOS are Occupational Standards which apply uniquely in the Indian context
Occupation	Occupation is a set of job roles, which perform similar/related set of functions in an industry.
Organisational Context	Organisational Context includes the way the organization is structured and how it operates, including the extent of operative knowledge managers have of their relevant areas of responsibility.
Performance Criteria	Performance Criteria are statements that together specify the standard of performance required when carrying out a task.
Qualifications Pack(QP)	Qualifications Pack comprises the set of NOS, together with the educational, training and other criteria required to perform a job role. A Qualifications Pack is assigned a unique qualification pack code.
Qualifications Pack Code	Qualifications Pack Code is a unique reference code that identifies a qualifications pack.
Scope	Scope is the set of statements specifying the range of variables that an individual may have to deal with in carrying out the function which have a critical impact on the quality of performance required.
Sector	Sector is a conglomeration of different business operations having similar businesses and interests. It may also be defined as a distinct subset of the economy whose components share similar characteristics and interests.



Sub-Sector	Sub-sector is derived from a further breakdown based on the characteristics and interests of its components.
Sub-functions	Sub-functions are sub-activities essential to fulfil the achieving the objectives of the function.
Technical Knowledge	Technical Knowledge is the specific knowledge needed to accomplish specific designated responsibilities.
Unit Code	Unit Code is a unique identifier for a NOS unit, which can be denoted with an 'N'
Unit Title	Unit Title gives a clear overall statement about what the incumbent should be able to do.
Vertical	Vertical may exist within a sub-sector representing different domain areas or the client industries served by the industry.
Keywords /Terms	Description
NOS	National Occupational Standard(s)
NSQF	National Skills Qualifications Framework
OEM	Original Equipment Manufacturer
OS	Occupational Standard(s)
QP	Qualifications Pack
5 S	Technique of maintaining orderliness –Japanese terminology
CP	Control Plan
WI	Work Instructions



ISC/N0201: Carry out all enabling jobs pertaining to coke production



Overview

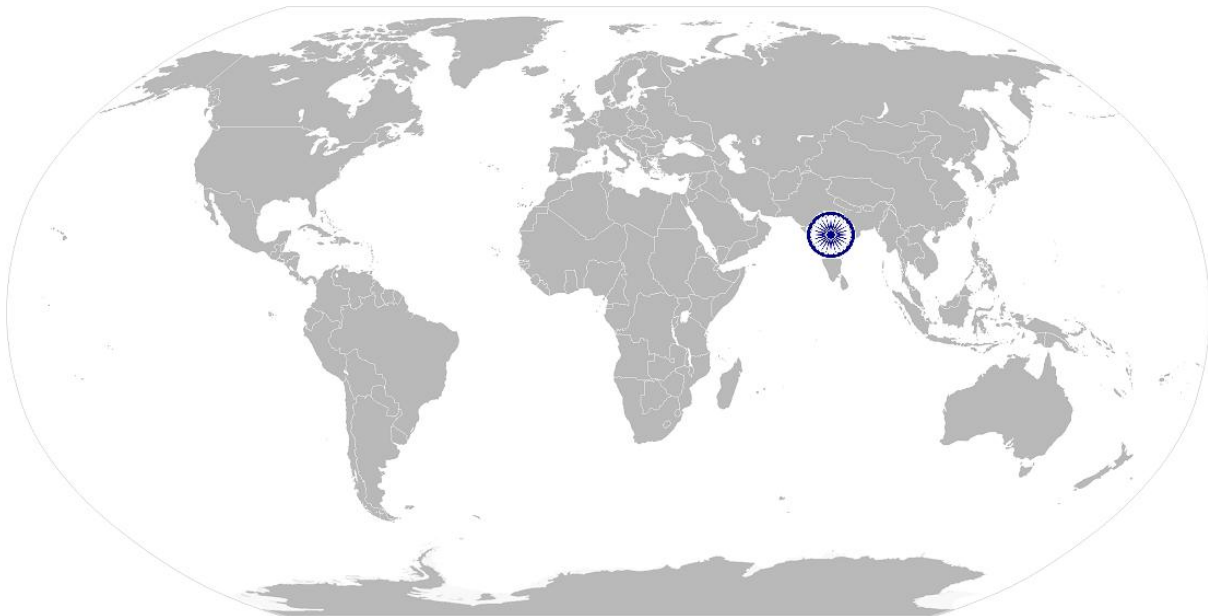
This NOS is about carrying out all enabling jobs pertaining to coke production



Unit Code	ISC/N0201
Unit Title (Task)	Carry out all enabling jobs pertaining to coke production
Description	This unit is about carrying out all enabling jobs pertaining to coke production.
Scope	This unit/task covers the following: - Understand all enabling jobs in accordance with job schedule - Understand the sequence of activities for execution - Seek clarification, if required - Understand advance preparation required for execution
Performance Criteria (PC) w.r.t. the Scope	
Element	Performance Criteria
Understand all enabling jobs in accordance with job schedule	To be competent, the user/individual on the job must be able to: PC1. Interpret the job of coke production and understand the job requirements PC2. Understand the specified enabling jobs for coke production
Understand the sequence of activities for execution	To be competent, the user/individual on the job must be able to: PC3. Clean the area specified as per job requirement PC4. Clean the battery machines as per machine cleaning schedule PC5. Identify ovens for technological cleaning of gas carrying equipments PC6. Carry out technological cleaning as per oven identified PC7. Operate designated cleaning equipment of relevance PC8. Seal the cleaning surfaces and regulate them as necessary, if required
Seek clarification, if required	To be competent, the user/individual on the job must be able to: PC9. Identify any clarifications that he wants to seek with respect to the given schedule PC10. Recognize whom to contact for clarifications on the given schedule PC11. Escalate the concern to the supervisor or shift-in-charge, if needed
Understand advance preparation required for execution	To be competent, the user/individual on the job must be able to: PC12. Carry out preparatory cleaning job as per requirement PC13. Clean the area around quenching track, guide car track etc. PC14. Identify tools, tackles & supporting equipment required to perform the operation PC15. Ask helper to carry tools required to the desired work site PC16. Report to stores / supervisor in case of non-availability of tools & tackles or stock-out
Element	Knowledge and Understanding
A. Organisational	The user/individual on the job needs to know and understand:



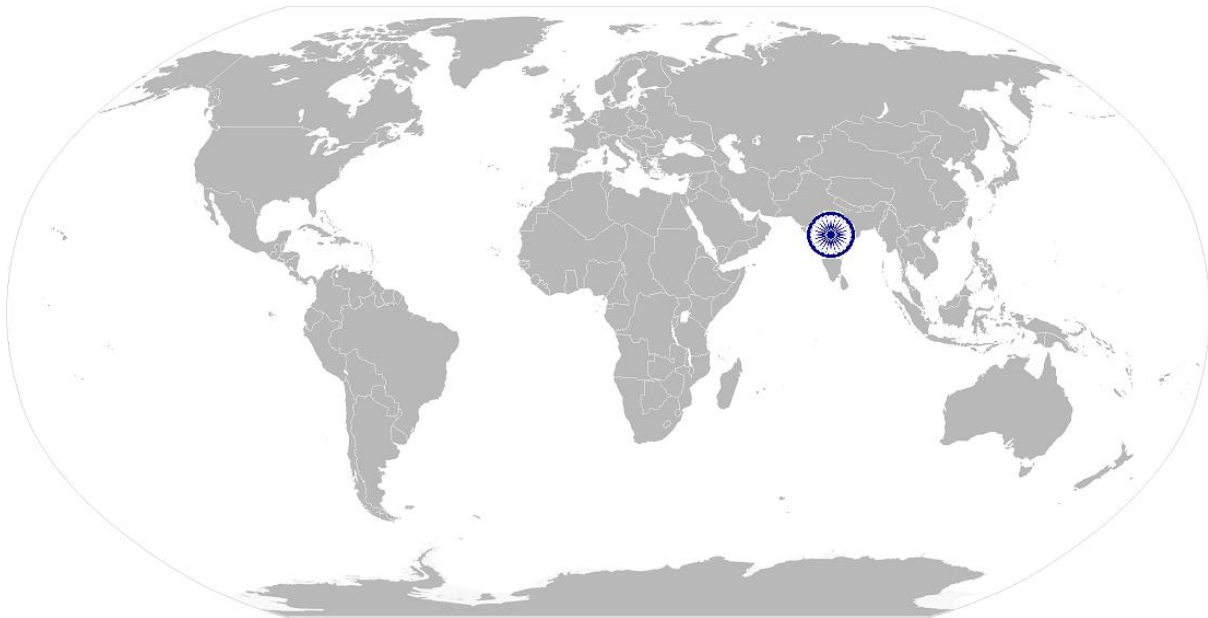
Context (Knowledge of the Company/ Organisation and its processes)	KA1. Quality and damage checks to be done and importance of the same KA2. Risk and impact of not following defined procedures/work instructions KA3. Escalation matrix for reporting identified issues
B. Technical Knowledge	The user/individual on the job needs to know and understand: KB1. How to switch on / off main with respect to Oven and Gas Collecting Main KB2. Tools & tackles to be used for the job KB3. How to operate the cleaning equipments KB4. How to to operate the pollution control equipments and systems e.g. HPLA system KB5. Normal running characteristics of relevant equipment KB6. Machine interlocking system e.g. interlocking with long travel of machine KB7. Implications of not adhering to sequence of activities and operations
Skills (S) w.r.t. the scope	
Element	Skills
A. Core Skills/ Generic Skills	Writing Skills
	The user/ individual on the job needs to know and understand how to: SA1. Construct simple sentences and express ideas clearly through written communication SA2. Fill up appropriate technical forms, process charts, activity logs in required format of the company
	Reading and Understanding Skills
	The user/individual on the job needs to know and understand how to: SA3. Read and interpret engineering and machine drawings w.r.t. coke oven battery and equipments SA4. Read and understand manuals, health and safety instructions, memos, reports, job cards, etc.
	Oral Communication (Listening and Speaking skills)
B. Professional Skills	The user/individual on the job needs to know and understand how to: SA5. Express statements, opinions or information clearly so that others can hear and understand SA6. Respond appropriately to queries SA7. Communicate with team members and supervisor
	Analytical Thinking & Problem Solving
	The user/individual on the job needs to know and understand how to: SB1. Diagnose common problems in the tools based on visual inspection, sound, temperature etc. SB2. Suggest improvements(if any) in operation processes based on experience





NOS Version Control

NOS Code	ISC/N0201		
Credits(NSQF)	TBD	Version number	1.0
Industry	Iron and Steel	Drafted on	08/09/2014
Industry Sub-sector	Steel	Last reviewed on	25/03/2015
Occupation	Coke Making	Next review date	25/03/2016





ISC/N0202: Prepare and carry out charging operation



National Occupational Standards

Overview

This unit is about preparing and carrying out charging operation in coke oven battery.



Unit Code	ISC/N0202
Unit Title (Task)	Prepare and carry out charging operation
Description	This unit is about preparing and carrying out charging operation in coke oven battery.
Scope	This unit/task covers the following: - Understand the charging schedule to plan the sequence of activities - Carry out the preparation activities top/stamped charging of oven - Carry out activities to charge and normalise the oven after completion of top/stamped charging
Performance Criteria (PC) w.r.t. the Scope	
Element	Performance Criteria
Understand the charging schedule to plan the sequence of activities	To be competent, the user/individual on the job must be able to: PC1. Understand the charging schedule PC2. Identify the oven for top/stamped charging PC3. Follow the planned sequence of activities for top/stamped charging operation
Carry out the preparation activities top/stamped charging of oven	To be competent, the user/individual on the job must be able to: PC4. Ensure free flow of charge coal into the charging car from coal tower bunker PC5. Inspect gas off take system for free gas flow after top/stamped charging of oven PC6. Inspect charging holes for free flow of charge coal for top charging of oven PC7. Connect the oven with the gas collecting main by application of HPLA system
Carry out activities to charge and normalise the oven after completion of top/stamped charging	To be competent, the user/individual on the job must be able to: PC8. Ensure top/stamped charging to the prepared oven PC9. Seal the charging hole lids for top charging by luting (various binding substances & mortar) PC10. Disconnect HPLA system to normalise the oven
Element	Knowledge and Understanding
A. Organisational Context (Knowledge of the Company/ Organisation and its processes)	The user/individual on the job needs to know and understand: KA1. Quality and damage checks to be done and importance of the same KA2. Risk and impact of not following defined procedures/work instructions KA3. Escalation matrix for reporting identified issues
B. Technical	The user/individual on the job needs to know and understand:

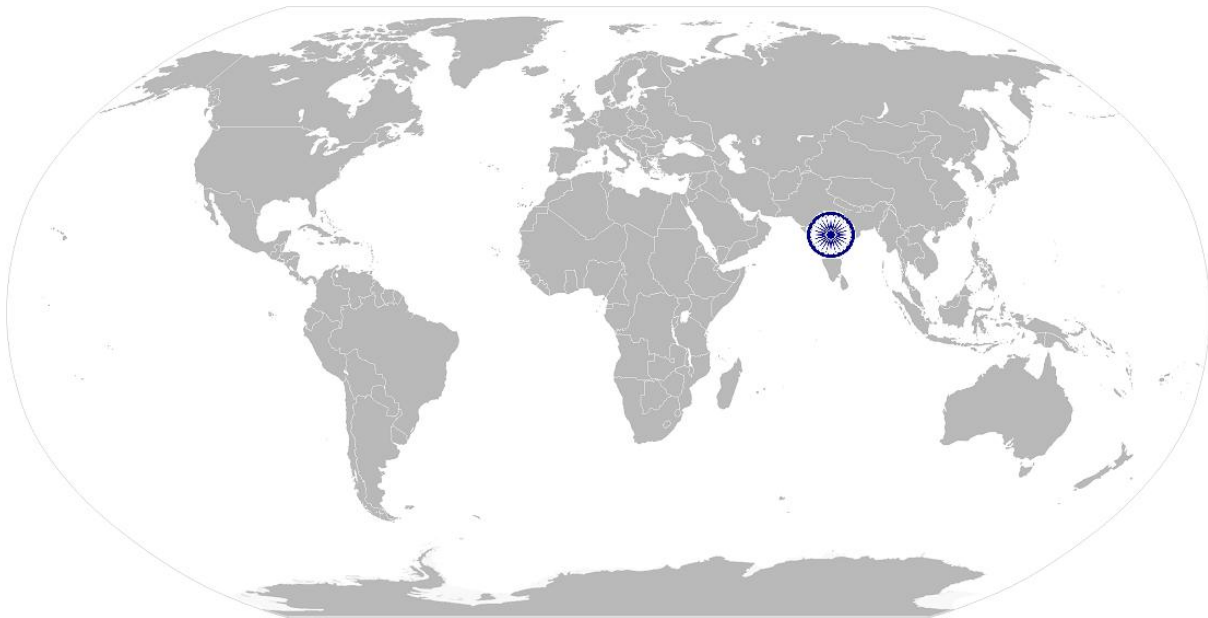


Knowledge	KB1. Normal running characteristics of relevant equipment KB2. How to switch on and off main with respect to oven and Gas Collecting Main KB3. Possible causes of operational problems e.g. unclean gas passage, oven pending pushing, oven specific repair work etc. KB4. Mechanised cleaning equipment KB5. Relevant pollution norms e.g. percentage leakage of lids (PLL), percentage leakage through off-take system (PLO) and charging emission KB6. Various interlocking system of charging car KB7: Implications due to failure of power, exhauster and Ammoniacal liquor supply system KB8. Implications of not adhering to sequence of activities and operations
Skills (S) w.r.t. the scope	
Element	Skills
A. Core Skills/ Generic Skills	Writing Skills
	The user/ individual on the job needs to know and understand how to:
	SA1. Construct simple sentences and express ideas clearly through written communication
	SA2. Fill up appropriate technical forms, activity logs in required format of the company
	Reading and Understanding Skills
	The user/individual on the job needs to know and understand how to:
B. Professional Skills	SA3. Read and understand manuals, health and safety instructions, memos, reports, job cards, specifications of spare parts etc.
	Oral Communication (Listening and Speaking skills)
	The user/individual on the job needs to know and understand how to:
	SA4. Express statements, opinions or information clearly so that others can hear and understand
	SA5. Respond appropriately to queries
	SA6. Communicate with supervisor, team members, other departments e.g. – stores, operations etc.
B. Professional Skills	Analytical Thinking & Problem Solving
	The user/individual on the job needs to know and understand how to:
	SB1. Diagnose common problems in the tools based on visual inspection, sound, temperature etc. SB2. Suggest improvements(if any) in assembly process based on experience



NOS Version Control

NOS Code	ISC/N0202		
Credits(NSQF)	TBD	Version number	1.0
Industry	Iron and Steel	Drafted on	08/09/2014
Industry Sub-sector	Steel	Last reviewed on	25/03/2015
Occupation	Coke Making	Next review date	25/03/2016





ISC/N0203: Prepare and carry out pushing operation



National Occupational Standards

Overview

This unit is about preparing and carrying out pushing operation



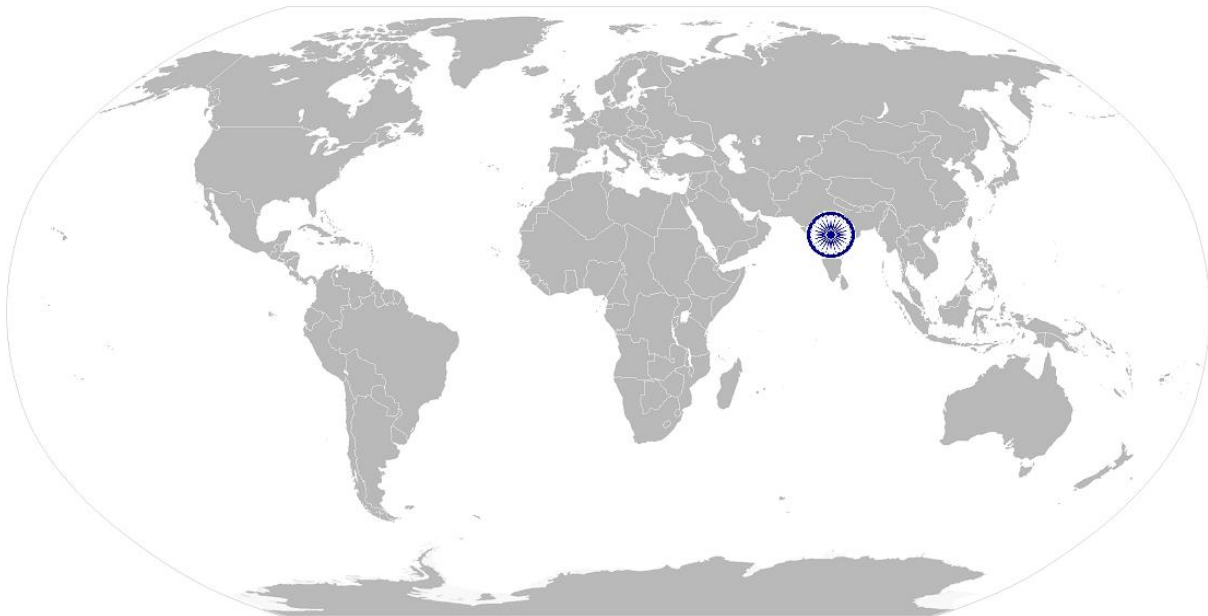
Unit Code	ISC/N0203
Unit Title (Task)	Prepare and carry out pushing operation
Description	This unit is about preparing and carrying out the sequence of activities for pushing of coke in coke oven battery.
Scope	This unit/task covers the following: - Understand the pushing schedule to plan the sequence of activities - Carry out the activities to prepare for pushing coke through oven - Carry out boxing-up of the oven doors - Carry out levelling of oven and normalization
Performance Criteria (PC) w.r.t. the Scope	
Element	Performance Criteria
Understanding the pushing schedule to plan the sequence of activities	To be competent, the user/individual on the job must be able to: PC1. Check the pushing schedule and begin with pushing activities PC2. Identify the oven doors for pushing PC3. Follow the planned sequence of activities for pushing operation PC4. Understand cross battery interlock and physical verification.
Carry out the activities to prepare for pushing coke through oven	To be competent, the user/individual on the job must be able to: PC5. Ensure isolation with Gas Collecting Main of identified oven PC6. Ensure to open the oven doors for pushing till completion of pushing
Carry out boxing-up of the oven doors	To be competent, the user/individual on the job must be able to: PC7. Clean sealing surfaces of oven door and door frame manually / mechanised cleaning equipment PC8. Regulate the oven door by adjusting the door regulation bolts (post charging and levelling operation)
Carry out levelling of oven and normalization	To be competent, the user/individual on the job must be able to: PC9. Clean the sealing surfaces of levelling window of pusher side door manually/mechanised equipment after levelling operation PC10. Ensure locking of levelling window on pusher side door PC11. Regulate the levelling window by regulation bolts PC12. Ensure quantity of coke after pushing (including uniform receiving)
Element	Knowledge and Understanding
A. Organisational Context	The user/individual on the job needs to know and understand:



(Knowledge of the Company/ Organisation and its processes)	KA1. Escalation matrix for reporting identified issues KA2. Risk and impact of not following defined procedures/work instructions
B. Technical Knowledge	The user/individual on the job needs to know and understand: KB1. Normal running characteristics of relevant equipment KB2. How to switch on and of main with respect to oven and Gas Collecting Main KB3. Mechanised cleaning equipment KB5. Relevant pollution norms e.g. Percentage Leakage Doors (PLD), KB6. Various interlocking system of pusher, guide car, set for stamped charge KB7: Implications due to failure of power, exhauster and Ammoniacal liquor supply system KB8. Implications of not adhering to sequence of activities and operations KB9. Response to emergencies e.g. Power failures, exhauster failure, leaker failure and fire
Skills (S) w.r.t. the scope	
Element	Skills
A. Core Skills/ Generic Skills	Writing Skills The user/ individual on the job needs to know and understand how to: SA1. Construct simple sentences and express ideas clearly through written communication SA2. Fill up appropriate technical forms, process charts, activity logs in required format of the company Reading and Understanding Skills The user/individual on the job needs to know and understand how to: SA3. Read and interpret engineering and machine drawings SA4. Read and understand manuals, memos, reports, job cards, etc. Oral Communication (Listening and Speaking skills) The user/individual on the job needs to know and understand how to: SA5. Express statements, opinions or information clearly so that others can hear and understand SA6. Respond appropriately to queries SA7. Communicate with supervisor, team members, other departments e.g. operations, stores etc.
B. Professional Skills	Analytical Thinking The user/individual on the job needs to know and understand how to: SB1. Diagnose common problems in the tools based on visual inspection, sound,



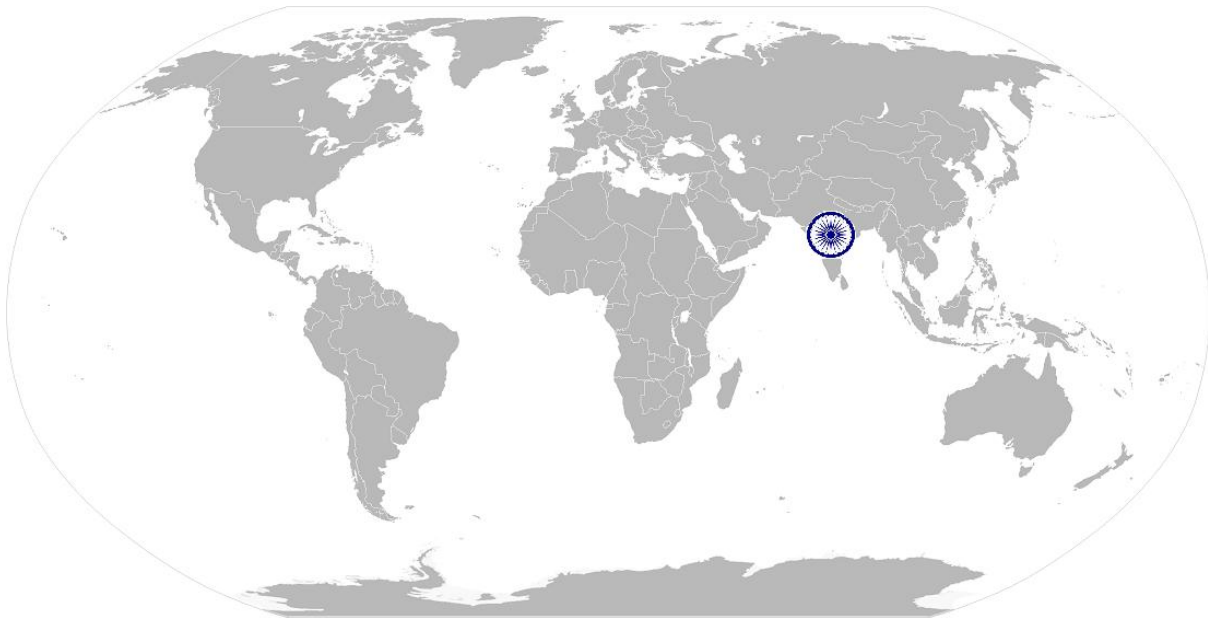
	temperature etc. SB2. Suggest improvements(if any) in process based on experience
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NOS Version Control

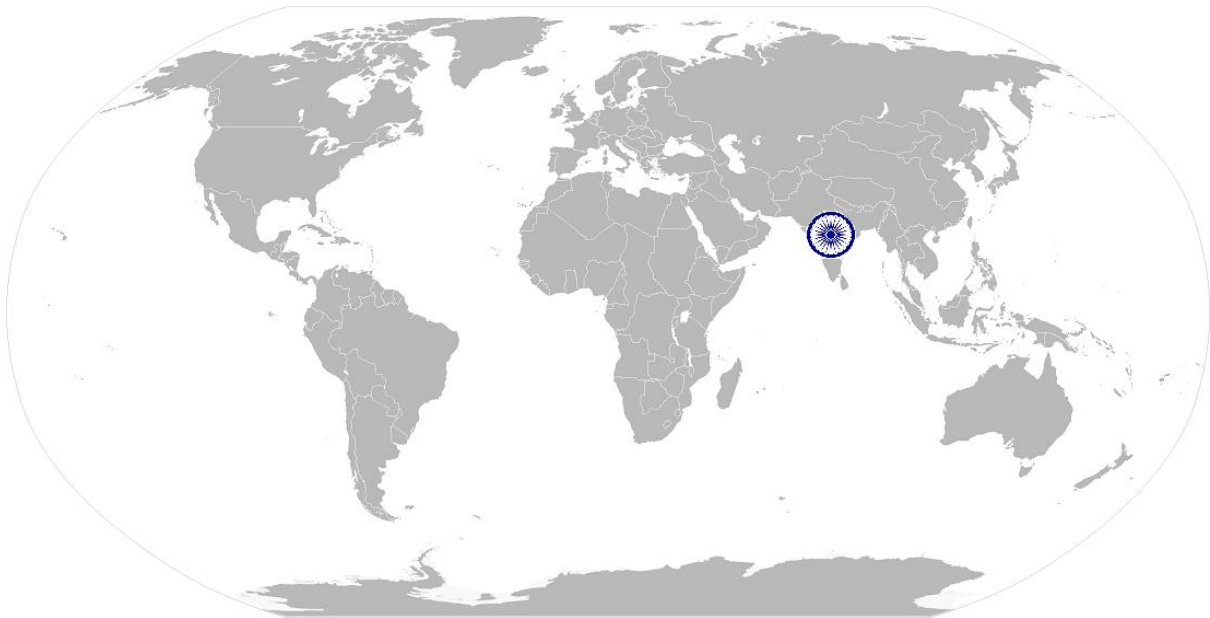
NOS Code	ISC/N0203		
Credits(NSQF)	TBD	Version number	1.0
Industry	Iron and Steel	Drafted on	08/09/2014
Industry Sub-sector	Steel	Last reviewed on	25/03/2015
Occupation	Coke Making	Next review date	25/03/2016





ISC/N0008: Use basic health and safety practices at the workplace

National Occupational Standards



Overview

This unit covers health, safety and security at the workplace. This includes procedures and practices that candidates need to follow to help maintain a healthy, safe and secure work environment.



Unit Code	ISC/N0008
Unit Title (Task)	Use basic health and safety practices at the workplace
Description	This OS unit is about knowledge and practices relating to health, safety and security that candidates need to use in the workplace. It covers responsibilities towards self, others, assets and the environment. It includes understanding of risks and hazards in the workplace, along with common techniques to minimize risk, deal with accidents, emergencies, etc.
Scope	This unit/task covers the following: • Health and safety procedures • Fire safety procedures • Emergencies, rescue and first aid procedures
Performance Criteria (PC) w.r.t. the Scope	
Element	Performance Criteria
Health and safety procedures	The user/individual on the job should be able to: PC1. Use protective clothing/equipment for specific tasks and work conditions Protective clothing includes: • Leather or asbestos gloves • Flame proof aprons • Flame proof overalls buttoned to neck • Cuff less (without folds) trousers • Reinforced footwear • Helmets/hard hats • Cap and shoulder covers • Ear defenders/plugs • Safety boots • Knee pads • Particle masks • Glasses/gloves/visors Equipment includes: • Hand shields • Machine guards • Residual current devices • Shields • Dust sheets • Respirator PC2. State the name and location of people responsible for health and safety in the

	workplace Various areas are listed below: • On chemical containers • Equipment • Packages • Inside buildings • Open areas, public places etc. PC3. State the names and location of documents that refer to health and safety in the workplace PC4. Identify job-site hazardous work and state possible causes of risk or accident in the workplace Hazards include: • Working with electrical and thermal tools and equipment • Sharp edged and heavy tools • Heated metals • Oxyfuel and gas cylinders • Welding radiation • Surfaces: sharp, slippery, uneven, chipped, broken, etc. • Substances: chemicals, gas, oxy-fuel, fumes, dust, etc. • Physical: working at heights, large and heavy objects and machines, sharp and piercing objects, tolls and machines, intense light, load noise, obstructions in corridors, by doors, blind turns, noise, over stacked shelves and packages, etc. • Electrical: power supply and points, loose and naked cables and wires, electrical machines and appliances, etc. PC5. Carry out safe working practices while dealing with hazards to ensure the safety of self and others state methods of accident prevention in the work environment of the job role Safe working practices include: • Using protective clothing and equipment • Putting up and reading safety signs • Handle tools in the correct manner and store and maintain them properly • Keep work area clear of clutter, spillage and unsafe object lying casually • While working with electricity take all electrical precautions like insulated clothing, adequate equipment insulation, use of control equipment, dry work area, switch off the power supply when not required, etc. • Safe lifting and carrying practices • Use equipment that is working properly and is well maintained • Take due measures for safety while working in confined places, trenches or at heights, etc. Including safety harness, fall arrestors etc. Methods are: • Training in health and safety procedures • Using health and safety procedures • Use of equipment and working practices (such as safe carrying procedures) • Safety notices, advice
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	• Instruction from colleagues and supervisors PC6. State location of general health and safety equipment in the workplace PC7. Inspect for faults, set up and safely use steps and ladders in general use Faults : • Corrosion of metal components • Deterioration • Splits and cracks timber components • Imbalance • Loose rungs • Nuts or bolts, etc. Set up: • Firm/level base • Clip/lash down • Leaning at the correct angle, etc. PC8. Work safely in and around trenches, elevated places and confined areas PC9. Lift heavy objects safely using correct procedures PC10. Apply good housekeeping practices at all times. Good housekeeping practices: • Clean/tidy work areas • Removal/disposal of waste products • Protect surfaces PC11. Identify common hazard signs displayed in various areas PC12. Retrieve and/or point out documents that refer to health and safety in the workplace
Fire safety procedures	The user/individual on the job should be able to: PC13. Use the various appropriate fire extinguishers on different types of fires correctly. Fire extinguishers: • Sand • Water • Foam • Co2 • Dry powder Fires: • Class A: Ordinary solid combustibles, e.g. wood, paper, cloth, plastic, charcoal etc. • Class B: Flammable liquids and gases, e.g. gasoline, propane, diesel fuel, tar, cooking oil and similar substances • Class C: Electrical equipment e.g. appliances, wiring, breaker panels etc. (these categories of fires become Class A, B, and D fires when the electrical equipment that initiated the fire is no longer receiving electricity) • Class D: Combustible metals such as magnesium, titanium, and sodium (these fires burn at extremely high temperatures and require special suppression agents) Causes of fires:



	• Heating of metal • Spontaneous ignition • Sparking, • Electrical heating • Loose fires (e.g. Smoking, welding, etc.) • Chemical fires, etc. PC14. Demonstrate rescue techniques applied during fire hazard PC15. Demonstrate good housekeeping in order to prevent fire hazards PC16. Demonstrate the correct use of a fire extinguisher
Emergencies, rescue and first-aid procedures	The user/individual on the job should be able to: PC17. Demonstrate how to free a person from electrocution PC18. Administer appropriate first aid to victims as required e.g. in case of bleeding, burns, choking, electric shock, poisoning etc. PC19. Demonstrate basic techniques of bandaging PC20. Respond promptly and appropriately to an accident situation or medical emergency in real or simulated environments. few General health and safety equipment are mentioned below : • Fire extinguishers • First aid equipment • Safety instruments and clothing • Safety installations, e.g. Fire exits, exhaust fans etc. PC21. Perform and organize loss minimization or rescue activity during an accident in real or simulated environments PC22. Administer first aid to victims in case of a heart attack or cardiac arrest due to electric shock, before the arrival of emergency services in real or simulated cases PC23. Demonstrate the artificial respiration and the CPR Process PC24. Participate in emergency procedures. Emergency procedures are: • Raising alarm • Safe/efficient evacuation • Correct means of escape • Correct assembly point • Roll call • Correct return to work PC25. Complete a written accident/incident report or dictate a report to another person, and send report to person responsible Incident Report should capture: • Name • Date/time of incident • Date/time of report, • Location • Environment conditions • Persons involved • Sequence of events • Injuries sustained • Damage sustained • Actions taken

	• Witnesses • Supervisor/manager notified Documents: • Fire notices • Accident reports • Safety instructions for equipment and procedures • Company notices and documents • Legal documents (e.g. Government notices) Job titles: • Health and safety officer • First aid officer • Fire officer PC26. Demonstrate correct method to move injured people and others during an emergency
Element	Knowledge and Understanding
A. Organisational Context (Knowledge of the Company/ Organisation and its processes)	The user/individual on the job needs to know and understand: KA1. State the names (and job titles if applicable), and describe where to find, all the people responsible for health and safety in a workplace KA2. State the names and location of documents that refer to health and safety in the workplace
B. Technical Knowledge	The user/individual on the job needs to know and understand: KB3. Meaning of “hazards” and “risks” KB4. Health and safety hazards commonly present in the work environment and related precautions KB5. Possible causes of risk, hazard or accident in the workplace and why risk and/or accidents are possible KB6. Activities and causes of risk and accident KB7. Methods of accident prevention KB8. Safe working practices when working with tools and machines KB9. Safe working practices while working at various hazardous sites KB10. Where to find all the general health and safety equipment in the workplace KB11. Various dangers associated with the use of electrical equipment KB12. Preventative and remedial actions to be taken in the case of exposure to toxic materials. • Exposure: ingested, contact with skin, inhaled • Preventative action: ventilation, masks, protective clothing/equipment • Remedial action: immediate first aid, report to supervisor • Materials: solvents, flux, lead KB13. Importance of using protective clothing/equipment while working KB14. Precautionary activities to prevent the fire accident Activities and causes: • Physical actions • Reading

	• Listening to and giving instructions • Inattention • Sickness and incapacity (e.g. Drunkenness) • Health hazards (e.g. Untreated injuries and contagious illness) KB15. Various causes of fire KB16. Techniques of using the different fire extinguishers KB17. Different methods of extinguishing fire KB18. Rescue techniques applied during a fire hazard KB19. Various types of safety signs and what they mean KB20. Appropriate basic first aid treatment relevant to the condition e.g. Shock, electrical shock, bleeding, breaks to bones, minor burns, resuscitation, poisoning, eye injuries KB21. Content of written accident report KB22. Potential injuries and ill health associated with incorrect manual handling KB23. Safe lifting and carrying practices KB24. Personal safety, health and dignity issues relating to the movement of a person by others KB25. Potential impact to a person who is moved incorrectly
Skills (S) w.r.t. the scope	
Element	Skills
A. Core Skills/ Generic Skills	Reading and Writing Skills
	The user/individual on the job needs to know and understand how to: SA1. Read and comprehend basic content to read labels, charts, signage's SA2. Read and comprehend basic English to read manuals of operations SA3. Read and write an accident/incident report in local language or English
	Oral Communication (Listening and Speaking skills)
	The user/individual on the job needs to know and understand how to: SA4. Question co-workers appropriately in order to clarify instructions and other issues SA5. Give clear instructions to co-workers, subordinates others
	Decision Making
	The user/individual on the job needs to know and understand how to: SA6. Make appropriate decisions pertaining to the concerned area of work with respect to intended work objective, span of authority, responsibility, laid down procedure and guidelines
B. Professional Skills	Plan and Organize
	The user/individual on the job needs to know and understand: SB1. Plan and organize their own work schedule, work area, tools, equipment and materials to maintain decorum and for improved productivity

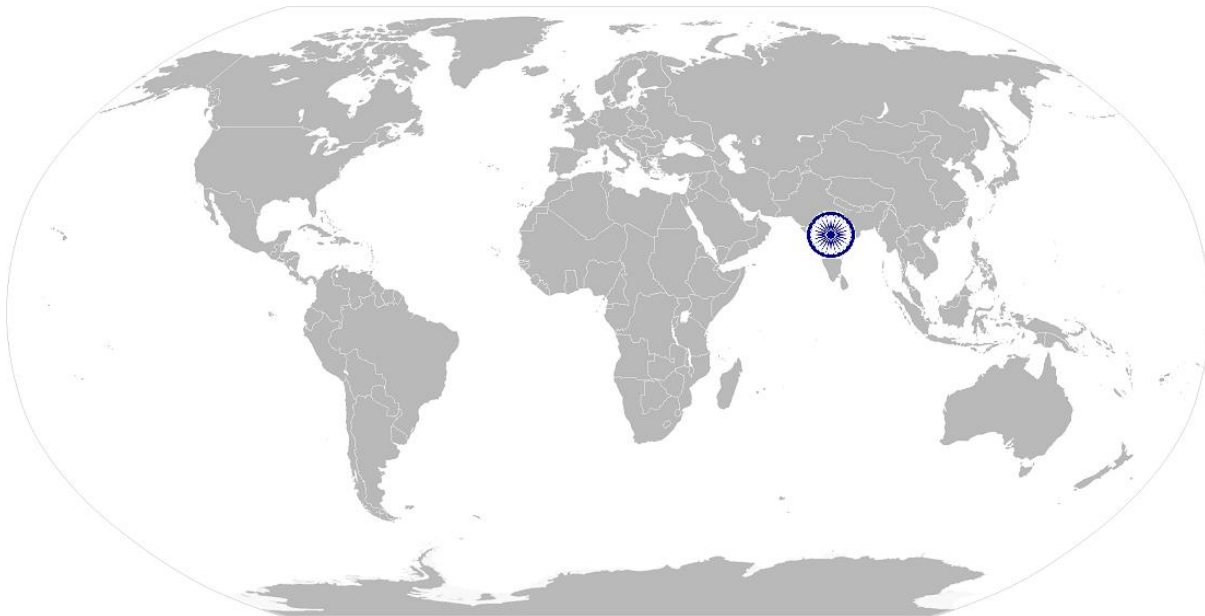


	Working with others
	The user/individual on the job needs to know and understand how to: SB2. Remain congenial while discussing and debating issues with co-workers SB3. Follow appropriate protocols for communication based on situation, hierarchy, organizational culture and practice SB4. Ask for, provide and receive required assistance where possible to ensure achievement of work related objectives SB5. Thank co-workers for any assistance received SB6. Offer appropriate respect based on mutuality and respect for fellow workmanship and authority
	Problem Solving
	The user/individual on the job needs to know and understand how to: SB7. Think through the problem, evaluate the possible solution(s) and suggest an optimum /best possible solution(s) SB8. Identify immediate or temporary solutions to resolve delays SB9. Identify sources of support that can be availed of for problem solving for various kind of problems SB10. Seek appropriate assistance from other sources to resolve problems SB11. Report problems that you cannot resolve to appropriate authority
	Analytical Thinking
	The user/individual on the job needs to know and understand how to: SB12. Identify cause and effect relations in their area of work SB13. Use cause and effect relations to anticipate potential problems and their solution



NOS Version Control

NOS Code	ISC/N0008		
Credits(NSQF)	TBD	Version number	1.0
Industry	Iron and Steel	Drafted on	23/07/2014
Industry Sub-sector	All Departments	Last reviewed on	30/12/2014
Occupation	Coke Making	Next review date	30/12/2015





ISC/N0096: Works effectively with others and manage team members

National Occupational Standards



Overview

This unit covers basic practices that improve effectiveness of working with others in an organisational set-up and managing team members



Unit Code	ISC/N0096
Unit Title (Task)	Works effectively with others and manage team members
Description	This unit covers basic etiquettes and competencies that a candidate is required to possess and demonstrate in their behaviour and interactions with others at the workplace. These cover areas such as communication etiquette, discipline, listening, handling conflict and grievances.
Scope	This unit/task covers the following: - Communicate effectively with team members, co-workers and supervisors - Encourage friendly working environment - Use appropriate titles and terms of respect - Follow code of conduct and ethics as applicable - Follow organisation specific policies - Demonstrate effective behaviours for team work - Ensure engagement of team members through on-the job handholding & support
Performance Criteria (PC) w.r.t. the Scope	
Element	Performance Criteria
Communicate effectively with co-workers and supervisors	The user/individual on the job should be able to: PC1. Accurately receive, absorb and share information and instructions from the supervisor and fellow workers, getting clarification where required PC2. Display appropriate communication etiquette while working PC3. Display active listening skills while interacting with others at work PC4. Use appropriate tone, pitch and language to convey politeness, assertiveness, care and professionalism
Demonstrate effective behaviours for team work	The user/individual on the job should be able to: PC5. Display helpful behaviour by assisting others in performing tasks in effective manner PC6. Consult with and assist others to maximize effectiveness and efficiency in carrying out tasks PC7. Demonstrate responsible and disciplined behaviours at the workplace PC8. Escalate grievances and problems to superiors
Ensure engagement of team members through on-the job handholding & support	The user/individual on the job should be able to: PC9. Communicate day-to-day objectives, instructions etc. to team members PC10. Guide the team members to manage day-to-day issues at work PC11. Gather concerns, feedback from team members and convey them to appropriate authorities



	PC12. Escalate grievances and problems to superiors
Element	Knowledge and Understanding
A. Organisational Context (Knowledge of the Company/ Organisation and its processes)	The user/individual on the job needs to know and understand: KA1. Legislation, standards, policies, and procedures followed in the company relevant to own employment and performance conditions KA2. Reporting structure, inter-dependent functions, lines and procedures in the work area KA3. Relevant people and their responsibilities within the work area KA4. Escalation matrix and procedures for reporting work and employment related issues
B. Technical Knowledge	The user/individual on the job needs to know and understand: KB1. Various categories of people that one is required to communicate and co-ordinate with in the organization KB2. Importance of teamwork in organizational and individual success KB3. Various components of effective communication and importance of effective communication while communicating professional circles KB4. Key elements of active listening and importance of active listening KB5. Importance of ethics for professional success KB6. Importance of discipline and disciplined behaviour for professional success KB7. Common reasons, importance and ways of managing interpersonal conflict effectively KB8. Expressing and addressing grievances appropriately and effectively
Skills (S) w.r.t. the scope	
Element	Skills
A. Core Skills/ Generic Skills	Writing skills and reading skills The user/individual on the job needs to know and understand how to: SA1. Read and comprehend basic content to read instructions, labels, charts, signage's SA2. Write notes in log books etc. SA3. Prepare incident reports, as appropriate
	Oral Communication (Listening and Speaking skills) The user/individual on the job needs to know and understand how to: SA4. Question co-workers appropriately in order to clarify instructions and other issues SA5. Communicate with co-workers SA6. Share information with Supervisors
	Decision Making

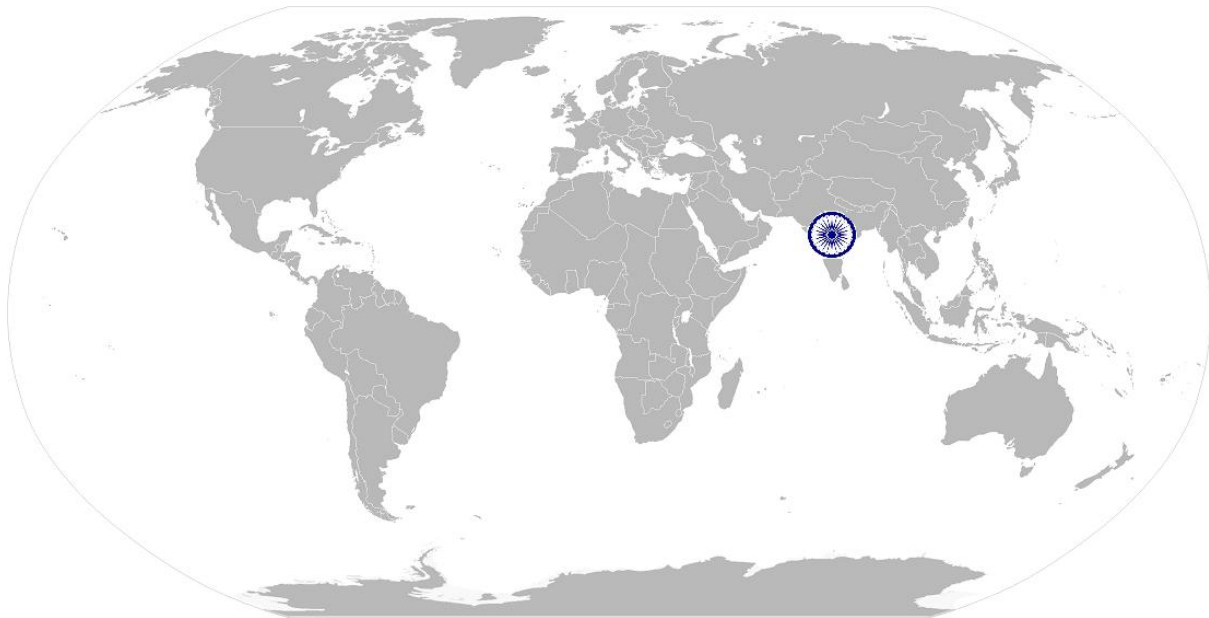


	The user/individual on the job needs to know and understand how to: SA7. Make appropriate decisions pertaining to the concerned area of work with respect to safely achieving work objective following the laid down procedure and guidelines
B. Professional Skills	Plan and Organize
	The user/individual on the job needs to know and understand: SB1. Plan and organize their own work schedule, work area, tools, equipment and materials to maintain decorum and for improved productivity
	Problem Solving
	The user/individual on the job needs to know and understand how to: SB2. Identify sources of support that can be availed of for problem solving for various kind of problems SB3. Seek appropriate assistance from other sources to resolve problems SB4. Report problems that cannot be resolved to appropriate authority
	Analytical Thinking
	The user/individual on the job needs to know and understand how to: SB5. Identify cause and effect relations in own area of work SB6. Use cause and effect relations to anticipate potential problems and their solution



NOS Version Control

NOS Code	ISC/N0096		
Credits(NSQF)	TBD	Version number	1.0
Industry	Iron and Steel	Drafted on	23/07/2014
Industry Sub-sector	All Departments	Last reviewed on	30/12/2014
Occupation	Coke Making	Next review date	30/12/2015





CRITERIA FOR ASSESSMENT OF TRAINEES

Job Role: Battery Operator

Qualification Pack: ISC/Q0201

Sector Skill Council: Indian Iron & Steel Sector Skill Council

Guidelines for Assessment:

1. Criteria for assessment for each Qualification Pack will be created by the Sector Skill Council. Each Performance Criteria (PC) will be assigned marks proportional to its importance in NOS. SSC will also lay down proportion of marks for Theory and Skills Practical for each PC.
2. The assessment for the theory part will be based on knowledge bank of questions created by the SSC.
3. Individual assessment agencies will create unique question papers for theory part for each candidate at each examination/training center (as per assessment criteria below)
4. Individual assessment agencies will create unique evaluations for skill practical for every student at each examination/training center based on this criteria.
5. To pass the Qualification Pack, every trainee should score a minimum of 60% in every NOS.
6. In case of successfully passing only certain number of NOS's, the trainee is eligible to take subsequent assessment on the balance NOS's to pass the Qualification Pack.

NOSs	PCs	Marks Allocated			
		Total Marks 1000	Out Of	Theory	Practical
ISC/N0201: Carry out all enabling jobs pertaining to coke production	PC1. Interpret the job of coke production and understand the job requirements	250	15	10	5
	PC2. Understand the specified enabling jobs for coke production		15	10	5
	PC3. Clean the area specified as per job requirement		15	5	10
	PC4. Clean the battery machines as per machine cleaning schedule		15	5	10
	PC5. Identify ovens for technological cleaning of gas carrying equipments		20	5	15
	PC6. Carry out technological cleaning as per oven identified		20	5	15
	PC7. Operate designated cleaning equipment of relevance		15	5	10



	PC8. Seal the cleaning surfaces and regulate them as necessary, if required		15	5	10
	PC9. Identify any clarifications that he wants to seek with respect to the given schedule		15	5	10
	PC10. Recognize whom to contact for clarifications on the given schedule		15	5	10
	PC11. Escalate the concern to the supervisor or shift-in-charge, if needed		15	5	10
	PC12. Carry out preparatory cleaning job as per requirement		15	5	10
	PC13. Clean the area around quenching track, guide car track etc.		15	5	10
	PC14. Identify tools, tackles & supporting equipment required to perform the operation		15	5	10
	PC15. Ask helper to carry tools required to the desired work site		15	5	10
	PC16. Report to stores / supervisor in case of non-availability of tools & tackles or stock-out		15	5	10
		Total	250	90	160
ISC/N0202: Prepare and carry out charging operation	PC1. Understand the charging schedule	250	25	10	15
	PC2. Identify the oven for top/stamped charging		25	10	15
	PC3. Follow the planned sequence of activities for top/stamped charging operation		25	10	15
	PC4. Ensure free flow of charge coal into the charging car from coal tower bunker		25	10	15
	PC5. Inspect gas off take system for free gas flow after top/stamped charging of oven		25	10	15
	PC6. Inspect charging holes for free flow of charge coal for top charging of oven		25	10	15
	PC7. Connect the oven with the gas collecting main by application of HPLA system		25	10	15
	PC8. Ensure top/stamped charging to the prepared oven		25	10	15



	PC9. Seal the charging hole lids for top charging by luting (various binding substances & mortar)		25	10	15
	PC10. Disconnect HPLA system to normalise the oven		25	10	15
		Total	250	100	150
ISC/N0203: Prepare and carry out pushing operation	PC1. Check the pushing schedule and begin with pushing activities	250	25	10	15
	PC2. Identify the oven doors for pushing		25	10	15
	PC3. Follow the planned sequence of activities for pushing operation		25	10	15
	PC4. Understand cross battery interlock and physical verification		25	10	15
	PC5. Ensure isolation with Gas Collecting Main of identified oven		25	10	15
	PC6. Ensure to open the oven doors for pushing till completion of pushing		25	10	15
	PC7. Clean sealing surfaces of oven door and door frame manually / mechanised cleaning equipment				
	PC8. Regulate the oven door by adjusting the door regulation bolts (post charging and levelling operation)		20	5	15
	PC9. Clean the sealing surfaces of levelling window of pusher side door manually/mechanised equipment after levelling operation		20	5	15
	PC10. Ensure locking of levelling window on pusher side door		20	5	15
	PC11. Regulate the levelling window by regulation bolts		20	5	15
	PC12. Ensure quantity of coke after pushing (including uniform receiving)		20	5	15
		Total	250	85	165
ISC/N0008: Use basic health and safety practices at the workplace	PC1. Use protective clothing/equipment for specific tasks and work conditions	150	10	5	5
	PC2. State the name and location of people responsible for health and safety in the workplace		5	0	5
	PC3. State the names and location of documents that refer to health and safety in the workplace		1	0	1



PC4. Identify job-site hazardous work and state possible causes of risk or accident in the workplace	9	5	4
PC5. Carry out safe working practices while dealing with hazards to ensure the safety of self and others state methods of accident prevention in the work environment of the job role	10	5	5
PC6. State location of general health and safety equipment in the workplace	5	0	5
PC7. Inspect for faults, set up and safely use steps and ladders in general use	5	0	5
PC8. Work safely in and around trenches, elevated places and confined areas	5	0	5
PC9. Lift heavy objects safely using correct procedures	5	0	5
PC10. Apply good housekeeping practices at all times	1	0	1
PC11. Identify common hazard signs displayed in various areas	6	5	1
PC12. Retrieve and/or point out documents that refer to health and safety in the workplace	4	0	4
PC13. Use the various appropriate fire extinguishers on different types of fires correctly	9	5	4
PC14. Demonstrate rescue techniques applied during fire hazard	10	5	5
PC15. Demonstrate good housekeeping in order to prevent fire hazards	1	0	1
PC16. Demonstrate the correct use of a fire extinguisher	4	0	4
PC17. Demonstrate how to free a person from electrocution	5	0	5
PC18. Administer appropriate first aid to victims as required e.g. in case of bleeding, burns, choking, electric shock, poisoning etc.	10	5	5
PC19. Demonstrate basic techniques of bandaging	5	0	5



	PC20. Respond promptly and appropriately to an accident situation or medical emergency in real or simulated environments		10	5	5
	PC21. Perform and organize loss minimization or rescue activity during an accident in real or simulated environments		5	0	5
	PC22. Administer first aid to victims in case of a heart attack or cardiac arrest due to electric shock, before the arrival of emergency services in real or simulated cases		5	0	5
	PC23. Demonstrate the artificial respiration and the CPR Process		5	0	5
	PC24. Participate in emergency procedures		5	0	5
	PC25. Complete a written accident/incident report or dictate a report to another person, and send report to person responsible		9	5	4
	PC26. Demonstrate correct method to move injured people and others during an emergency		1	0	1
		Total	150	45	105
ISC/N0096: Works effectively with others and manage team members	PC1. Accurately receive, absorb and share information and instructions from the supervisor and fellow workers, getting clarification where required	100	5	0	5
	PC2. Display appropriate communication etiquette while working		10	0	10
	PC3. Display active listening skills while interacting with others at work		10	0	10
	PC4. Use appropriate tone, pitch and language to convey politeness, assertiveness, care and professionalism		5	0	5
	PC5. Display helpful behaviour by assisting others in performing tasks in effective manner		10	0	10
	PC6. Consult with and assist others to maximize effectiveness and efficiency in carrying out tasks		10	0	10



PC7. Demonstrate responsible and disciplined behaviours at the workplace	5	0	5
PC8. Escalate grievances and problems to superiors	5	0	5
PC9. Communicate day-to-day objectives, instructions etc. to team members	15	5	10
PC10. Guide the team members to manage day-to-day issues at work	5	0	5
PC11. Gather concerns, feedback from team members and convey them to appropriate authorities	15	5	10
PC12. Escalate grievances and problems to superiors	5	0	5
Total	100	10	90

